AI-POWERED CYBERSECURITY

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Assessment AI-Powered Cybersecurity

Task A – Secure a Student Account

A1. Five distinct risks

Risk	Asset at risk	CIA impact
Reusing the same short password	Email and learning portal accounts	Confidentiality: if one account is compromised, an attacker may access both accounts.
Approving unexpected sign-in prompts	Learning portal account	Confidentiality: an unauthorised person may gain access after an unexpected approval.
Postponing device updates	Laptop and stored information	Integrity and availability: unpatched vulnerabilities may be exploited to alter data or disrupt the device.
Keeping only one project copy	Academic projects and files	Availability: laptop damage, theft or malware could cause permanent data loss.
Sharing a laptop while still signed in	Personal files and learning portal	Confidentiality and integrity: another user could view private information or modify/delete files.

A2. Practical controls, reasons and limitations

Risk	Control	Why it helps and limitation
Password reuse	Use a unique, strong password for each account.	Reduces the chance that one leaked password exposes multiple accounts. Limitation: weak recovery procedures can still create risk.
Unexpected prompts	Enable MFA and deny unfamiliar requests.	Adds another verification layer. Limitation: social engineering or accidental approval can still defeat MFA.
Delayed updates	Enable automatic security updates.	Fixes known vulnerabilities. Limitation: an update may occasionally cause compatibility problems.
Single project copy	Keep encrypted backups in cloud storage or on an external drive.	Supports recovery after data loss. Limitation: backups must be updated and protected.
Shared signed-in laptop	Lock the screen and create separate user accounts.	Reduces access to Mira's personal session. Limitation: physical access or administrator privileges may still create risks.

Priorities: (1) Secure passwords and enable MFA because compromised credentials can expose important accounts. (2) Stop approving unexpected prompts to prevent accidental access approval. (3) Back up important projects to protect academic work from device failure, ransomware or accidental deletion. Mira should then update her laptop and create separate accounts for people sharing it.

A3. Authentication versus authorisation

Authentication verifies a user's identity. For example, when Mira enters her username, password and OTP to access the learning portal, the system checks whether she is the genuine account holder. Authorisation determines what an authenticated user is allowed to access or do. Mira may be allowed to view marks, download study materials and submit assignments, but not modify faculty records or manage other students' accounts. In short, authentication verifies identity, while authorisation determines permissions.

Task B – Analyse a Suspicious Message

B1. Five warning signs

Warning sign / evidence	Why it needs scrutiny
"FINAL WARNING - scholarship expires in 20 minutes"	Creates urgency and pressures the student to act without checking the message.
From: awards@campus-awards.example	The sender domain differs from the college's normal campus.example domain and needs independent verification.
Reply-To: help@personal-awards.example	The reply address differs from the sender address and may redirect communication elsewhere.
"enter your college password and OTP"	A request for a password and OTP through an unverified link is highly suspicious and could expose account credentials.
"Do not contact the college office"	Discourages independent verification and attempts to isolate the student from official support.

Another warning sign is the request for a photograph of an identity card, which could expose sensitive personal information.

B2. Safe verification process

1. Do not click the link or reply to the suspicious message.
2. Open the college's official website by typing its known address manually or using a trusted bookmark.
3. Use the official student portal if necessary.
4. Contact the scholarship department through a phone number or email address obtained independently from an official college source.
5. Ask the college to confirm whether the scholarship announcement is genuine.
6. Report the message to the college IT or cybersecurity team.

HTTPS encrypts communication between a browser and a website, but it does not prove that the website is legitimate. Phishing websites can also use HTTPS. A familiar display name can be forged, so the actual sender domain, destination URL and independent official confirmation must be checked.

B3. Response plan if a student has entered a password and OTP

1. **Containment:** Contact the college IT team immediately and change the compromised password through the official portal using a trusted device.
2. **Other accounts:** If the same password was reused, change it on those accounts too.
3. **Sessions:** Ask authorised IT staff to revoke active sessions, invalidate suspicious tokens and temporarily block the account if needed.
4. **MFA:** Review MFA settings, remove unknown devices and report suspicious authentication activity.
5. **Reporting:** Report the incident to college IT and provide the suspicious email and relevant timestamps. Do not send anyone the password or OTP.
6. **Monitoring:** Check account activity, profile changes, email forwarding rules, assignment submissions and any unauthorised access or transactions.

B4. Fact that cannot be established

The message alone cannot establish whether it was sent by an authorised college representative or by an attacker using a forged identity. Additional evidence could include full email headers, SPF/DKIM/DMARC authentication results, domain ownership information and confirmation from the official college scholarship department.

Task C – Triage a Sign-in Alert

C1. Four-point timeline

Time (IST)	Observed event	Interpretation
09:00–09:01	Password success and MFA approved from 192.0.2.10.	Appears consistent with a normal login from the usual classroom egress IP.
10:14–10:15	Two password attempts failed from 198.51.100.23.	Could indicate attempts to access the account using incorrect credentials.
10:16–10:17	Password success, followed by an MFA denial from 198.51.100.23.	A correct password was accepted, but the first MFA request was denied. The logs do not show how the password was obtained.
10:18–10:20	MFA approved and session created. Student says they approved the prompt accidentally and did not initiate a login.	The approval may have enabled an unauthorised session; this requires investigation.

C2. Likely and alternative explanations

Likely explanation: An attacker may have obtained student07's password and attempted to access the account. After two failed attempts, password authentication succeeded. The attacker then triggered an MFA request. The student mistakenly approved the prompt at 10:18, allowing a session to be created.

Alternative explanation: The login could have been initiated by a legitimate application, a previously used device or another authorised user, while the different IP address resulted from a network change or VPN. This possibility does not fully explain the student's statement that they did not initiate the login.

Two additional checks: (1) Review device information, browser details, session records and available location information to see whether the session matches a known device. (2) Check authentication logs, MFA records and network activity to determine whether the login came from an unfamiliar device or application. An IP address alone cannot identify a person.

C3. Three prioritised response actions for authorised IT

1. **Contain:** Temporarily restrict the potentially compromised account, revoke active sessions and require a password reset. Review MFA settings and re-enrol MFA if needed.
2. **Preserve evidence:** Secure copies of authentication logs, MFA records, timestamps, session details and relevant network information. Maintain an investigation record and do not delete suspicious logs.
3. **Investigate and recover:** Examine login activity, identify possible unauthorised access and check whether account information or files were modified. Restore normal access only after securing the account and verifying its safety.

C4. Incident summary (60–100 words)

The sign-in logs show two failed password attempts followed by a successful password authentication from an unfamiliar IP address. The MFA request was initially denied but later approved at 10:18, after which a session was created. The student confirmed that they did not initiate the login but accidentally approved the MFA prompt. This indicates a possible account compromise through credential misuse and MFA approval manipulation. The source IP does not identify the attacker. The priority is to secure the account, revoke active sessions and preserve evidence for further investigation.

Task D – Review an AI Security Answer

D1. Five unsafe or unsupported claims/actions and corrections

AI claim/action	Correction
"The email is safe because it uses HTTPS."	HTTPS encrypts communication but does not prove a website is legitimate. Phishing websites can also use HTTPS.
"The sign-in logs prove that the person using 198.51.100.23 stole the password."	Logs show authentication activity from an IP address, not the identity of a person or how a password was obtained.
"MFA prevents every account compromise."	MFA adds protection, but social engineering, MFA fatigue and stolen sessions can still lead to compromise.
"Delete the suspicious logs."	Preserve suspicious logs because they are important evidence for incident investigation.
"Send the student's password to support so they can test it."	Never share passwords with support staff. Secure the account through an official password-reset process.

D2. Improved prompt for an AI tool

“Analyse only the fictional cybersecurity evidence provided in this assessment. Identify the observable facts separately from interpretations and assumptions. For each conclusion, explain which supplied evidence supports it and identify any uncertainty or missing information. Do not claim that an IP address identifies a person or that HTTPS proves a website is safe. Do not request, reveal or process real passwords, OTPs, identity documents or personal data. Recommend safe, proportionate actions for account protection, reporting and evidence preservation. If the evidence is insufficient to reach a conclusion, state what additional authorised evidence would be needed.”

D3. Corrected security advice (100–150 words)

Do not trust a scholarship message only because it displays a familiar college name or uses HTTPS. Avoid clicking unexpected links or sharing your password, OTP or identity documents through an unverified message. Verify scholarship announcements through the college's official website or by contacting the scholarship office using independently obtained contact details. If you have entered your credentials, immediately contact the college IT team, change your password through the official portal and request that active sessions be revoked. Review MFA settings and check for unfamiliar account activity. If the same password is used elsewhere, change it on those accounts too. Preserve the suspicious message and report it to the authorised IT team. Never delete relevant security logs or send passwords to support staff.

D4. AI-use declaration and evidence/reference notes

AI-use declaration: ChatGPT was used to assist with organising and drafting explanations for this assessment. The supplied fictional scenarios and logs were used as the basis for the answers. The content should be reviewed and edited by the student to ensure understanding and accuracy before submission.

Evidence/reference note 1 – CISA: The assessment provides CISA Secure Our World as reference reading for basic account-security and phishing guidance. It supports the use of strong account practices, caution with suspicious messages and independent verification.

Evidence/reference note 2 – NIST: The assessment provides the NIST Generative AI Profile (AI 600-1) as reference reading. It is relevant to checking AI-generated claims, recognising that AI outputs can be inaccurate, and verifying conclusions against supplied evidence.

References

1. Cybersecurity and Infrastructure Security Agency (CISA), *Secure Our World*. https://www.cisa.gov/secure-our-world
2. National Institute of Standards and Technology (NIST), *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile*, AI 600-1. https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf
3. EWB Courses, *AI-Powered Cybersecurity – Assessment 01: Security Foundations*, issued 18 October 2026.